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Research Proposal

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Research Topic	Balancing Privacy, Accuracy, and Fairness: A Comparative Study of Privacy-Preserving Techniques in Pedestrian Detection Models for Public Surveillance		
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Agreement

I S.A.G.N. Jayasekara, willing to submit the following Proposal format and I am confirming that this topic is a novel research and I am aware of the consequences occurs in the case of plagiarism found.

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Background

In this day and age, public surveillance and monitoring systems have become an integral part of modern infrastructure, helpful for applications such as pedestrian monitoring, traffic management, and enhancing public safety within urban environments [1]. These systems increasingly leverage advanced computer vision models for real-time detection and tracking of individuals [2], [3]. While offering larger benefits for urban planning and security [1], [4], their pervasive deployment raises substantial ethical and practical concerns regarding individual privacy, data security, and the potential for algorithmic bias [5].

The fundamental challenge is in balancing the operational utility of surveillance systems with the need to protect personal privacy [5]. The collection and processing of vast amounts of visual data inherently increase the risk of compromising personally identifiable information [5]. To mitigate these privacy risks and comply with regulatory frameworks, various privacy-preserving techniques have been developed. These include methods like blurring, pixelation, selective encryption, and full-body anonymization [6], [7]. However, implementing such techniques often introduces a trade-off, potentially degrading the accuracy and performance of the underlying detection and recognition [6], [7]. For instance, studies have shown that traditional image anonymization, especially full-body anonymization, can substantially impact model performance, although realistic anonymization, particularly for faces, might result in minimal performance degradation [8]. The intensity of privacy preservation directly influences this balance; stronger anonymization methods show a significant decrease in detection [6], [5].

Moreover, the fairness of computer vision systems can be affected by the application of privacy-preserving transformations [9]. The majority of privacy-saving algorithms have not been studied in terms of fairness, although the performance rates of the facial recognition systems with various phenotypes have been known to differ [10]. One should understand whether these techniques impose a biased influence on different demographic or contextual sub-populations that can cause or widen performance disparities. Such prejudices can cause variations in effects and discriminatory results.



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Therefore, a thorough analysis is needed to identify the impact of various privacy-friendly transformations and their corresponding levels of intensity on the accuracy of pedestrian detection, as well as the justification of the results in diverse communities [10], [11]. This study aims to identify the optimal trade-offs, thereby facilitating the design of responsible, privacy-conscious computer vision systems that strike a balance between individual privacy protection and model performance in the context of positive public surveillance [12].

Literature Review

Public surveillance systems have become very common in cities and are essential for monitoring pedestrians, controlling traffic, and improving the level of security of the citizens [1], [4]. Technology in computer vision, especially models that are built on deep learning, has enhanced real-time pedestrian detection and tracking, thus making cities smarter and their urban planning decisions more improved [3], [5]. However, these kinds of systems raise serious issues with the invasion of privacy, the biases of the algorithms, and the dilemma between the usefulness of the system and its ethical standards [5], [10].

This literature review tries to identify the existing literature gaps on privacy-preserving techniques in pedestrian detection in public surveillance, with a stronger focus on their impact on accuracy and fairness. This review relies on research papers that cover the detection methods, privacy mechanisms, performance degradation, amplification of biases, and the new frameworks that are developed to balance these aspects. This review identifies that the current methodological gaps include a lack of detailed assessment of diverse datasets and the need to have unified privacy-accuracy-fairness models.

Pedestrian Detection in Surveillance Systems

Modern surveillance systems rely heavily on pedestrian detection, which applies computer vision models to detect and track people in a crowded or large area [2]. Previous work has included the use of traditional algorithms like the histogram of oriented gradients (HOG) with support-vector machines (SVM), but the latest developments have been inclined towards deep convolutional neural networks (CNNs) due to their effectiveness [13], [14]. For instance, V. V. Molchanov et al. in the research paper [3] introduced a complete convolutional YOLO-based pedestrian detector in video surveillance with high precision in real-time scenarios through efficient processing of video frames. Also, H. Sui et al. in the research titled "FA-YOLO: A Pedestrian Detection Algorithm with Feature Enhancement and Adaptive Sparse Self-Attention" [2] presented an extended version of YOLO that added feature enhancement and adaptive sparse self-attention, and performed better on urban scenes with complex backgrounds.

High-quality datasets are essential in training pedestrian detection models to maintain accuracy and reduce errors. CityPersons dataset offers a wide range of city scenes with annotations of the pedestrian bounding boxes, addressing issues like occlusions, as well as different object sizes [14]. WiderPerson also expands this by including dense crowds of people in wild environments, thus improving the strength of model training to be used in real-world situations [15]. However, the datasets include personally identifiable information (PII), which increases the privacy threat when used in the context of general surveillance [16]. Research papers use computational intelligence to identify harmful pedestrian behavior in urban areas. The focus should be on creating proactive safety mechanisms that forecast pedestrian behavior without reducing the detection speed [4].

Despite all these technological developments, detection models are still vulnerable to environmental factors like light conditions, occlusion, and viewpoint alteration [17]. Similarly, in a study that examined data-augmented approaches, including geometric techniques (e.g., rotations) and photometric techniques (e.g., brightness tuning) for pedestrian detection with thermal images, and showed that they



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improve generalization in both indoor and outdoor scenarios, whilst maintaining privacy because non-visual data modalities are used [18].

Datasets and Benchmarks for Privacy-Preserving Pedestrian Detection

The performance of pedestrian detection models largely depends on the presence of diverse and carefully labelled data, and the balance between usefulness and privacy to ensure that they do not accidentally reveal personal identifiable information (PII) in training and testing is important [19]. Conventional data collections like CityPersons contain more than 5,000 high-resolution urban snapshots, with about 35,000 pedestrian annotations, with labels for adults, children, and occlusions [14]. The dataset is well-suited for surveillance research, but using real-world images raises valid privacy concerns. The WiderPerson dataset expands on earlier work by including dense crowds in natural scenes, with 13,382 images and over 40,000 annotations [15]. This research shows how effective training is possible and, at the same time, highlights the need for anonymization of protocols when working on public data.

Specialized benchmarks have emerged to address these privacy concerns regarding public datasets. For example, the PP4AV dataset offers 3,447 driving images with 3,895 face and 8,028 license plate annotations from European cities, which are designed explicitly for evaluating anonymization in autonomous driving scenarios [20]. It supports privacy-preserving evaluations by focusing on identifiable elements (e.g., pedestrian faces with size thresholds for recognizability) and includes fisheye views, achieving a baseline Average Precision (AP50) of about 76.22% for faces in normal images. This dataset enables benchmarking of de-identification techniques like blurring without compromising scene utility, revealing performance drops for small or occluded objects.

CrowdTrack introduces a benchmark for multiple pedestrian tracking in challenging real-world scenarios, comprising 33 video sequences with over 700,000 annotations and 5,185 trajectories [21]. It uses privacy-preserving processing by blocking identifiable information, emphasizing dense crowds, occlusions, and irregular motions in surveillance (e.g., malls, stations). Evaluations revealed high-end trackers like ByteTrack suffer higher ID switches, showing the need for robust privacy-integrated benchmarks.

Other datasets emphasize multi-modal or thermal data for inherent privacy benefits. The KAIST Multispectral Pedestrian Dataset integrates 95,000 visible and infrared images, reducing identifiability in low-light surveillance [22]. EuroCity Persons (ECP) provides 47,000 images across varied weather conditions, with detailed annotations for robust privacy evaluations, as noted in fairness studies [23], [24]. These benchmarks highlight the shift toward datasets that enable privacy-by-design, though challenges remain in standardizing privacy metrics across them.

Privacy Concerns and Preservation Techniques

The collection of vast visual data in surveillance systems inherently risks exposing personally identifiable information (PII), leading to ethical and regulatory challenges under frameworks like the General Data Protection Regulation (GDPR) [5], [25]. Privacy-preserving techniques aim to mitigate this by anonymizing data at the source or during processing. Traditional methods include blurring, pixelation, and scrambling [26], [27]. Dufaux et al. proposed scrambling for video surveillance, selectively obscuring regions to protect identities while preserving scene utility [26].

Modern approaches leverage deep learning for more sophisticated anonymization. A study shows the impact of image anonymization on CV training, finding that face-specific anonymization causes minimal performance degradation compared to full-body methods [8]. AI-based anonymization for industrial image data, preserving features like edges and textures essential for downstream tasks [7]. De Coninck et al. developed a privacy-preserving visual analysis framework using adversarial training to obfuscate videos without sensitive labels, enabling edge deployment where data is anonymized on-device before cloud transmission [28].



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Encryption-based methods offer stronger guarantees. A Vector Homomorphic Encryption (VHE) scheme for privacy-preserving pedestrian detection, allowing HOG (Histogram of Oriented Gradients) feature extraction over encrypted images with SVM classification [13]. A study by Tian et al. proposed robust motion detection in encrypted streaming video, supporting object tracking without decrypting data [29]. Federated learning has emerged for collaborative training without sharing raw data, applied to person localization in public camera systems, using encrypted embeddings to match individuals across sites while preserving privacy [30].

Audio-visual and multi-modal techniques further enhance privacy. Vector-quantized variational autoencoders (cycle-VQ-VAE) can be used to encode videos with extracted audio as noise, compressing data and resisting side-channel attacks [31]. Lee et al. developed a UAV system with deep learning-based face anonymization, anonymizing faces in real-time for aerial surveillance [32].

Top-view surveillance offers inherent privacy benefits by reducing identifiable features. Li and Lo [33] reviewed top-view videos for pedestrian detection and tracking, noting reduced occlusions and better privacy compared to side-view cameras, though a limited field of view remains a challenge.

Impact on Accuracy and Utility

Privacy-preserving techniques often introduce trade-offs with detection accuracy, as anonymization can degrade semantic information [6]. Research paper evaluation of face anonymization methods (e.g., blurring, latent diffusion) on CV tasks like keypoint detection and segmentation, revealing a clear privacy-utility trade-off, while heavy blurring can reduce mean average precision (mAP) by 20-30% in pedestrian detection tasks, while latent diffusion minimizes loss to ~3-5% for keypoint detection on datasets like COCO. Latent diffusion minimized accuracy loss while providing strong anonymization [34]. Video-to-Text Pedestrian Monitoring (VTPM), using CV and large language models to describe activities textually, preserving privacy with negligible impact on monitoring utility[6].

Quantitative studies highlight varying degradation levels. Block scrambling with DCNN-based face detection, achieving high encryption security but noting reduced precision in occluded scenes [27] combined. Introduction of real-time privacy-preserving surveillance using selective obfuscation, maintaining detection rates above 85% in dynamic environments[16]. A paper by Yaghini et al. proposed learnable anonymization for pedestrian images, preserving re-identification utility while removing PII, with experiments showing minimal mAP drops on datasets like Market-1501 [9].

In thermal imaging, from the paper by Fanfani et al. demonstrated that augmentation mitigates accuracy loss in privacy-friendly modalities, achieving F1-scores comparable to RGB-based models [18]. However, intensive anonymization (e.g., heavy blurring) can reduce mean average precision (mAP) by up to 20-30% [7], [8].

Fairness Issues in Privacy-Preserving Systems

Fairness in pedestrian detection ensures equitable performance across demographics, but privacy techniques can inadvertently amplify biases [9], [35]. Buolamwini et al. in their research exposed intersectional disparities in facial recognition, with higher errors for darker-skinned females, which is a concern extending to pedestrian models [36]. Fairness in visual privacy algorithms, finding that anonymization disproportionately affects subgroups (e.g., by phenotype), leading to biased outcomes in recognition tasks [10].

Khoshkdahan et al. introduced FAIR-PED, using CLIP models to evaluate fairness in pedestrian detection, revealing biases toward lateral views and certain attributes (e.g., bags), with Equal Opportunity Difference (EOD) metrics showing up to 4.65% disparities [24]. Testing fairness in autonomous driving systems, uncovering biases related to age and skin tone, exacerbated by anonymization that alters feature distributions [35].



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Yaghini et al. proposed impartial learning to optimize the Pareto frontier of fairness, privacy, and utility, using differential privacy (DP) to mitigate biases [9]. Differential Privacy (DP) in deep learning is helpful in preventing disparate impacts, but at the cost of utility [37]. Maris et al. in their research paper applied DP and k-anonymity to pedestrian data, showing preserved re-identification but degraded demographic predictions, highlighting the need for bias-aware anonymization [38].

When evaluating the trade-offs in CV between attribute privacy, fairness, and utility, advocating for frameworks that model these interdependencies empirically [12].

Trade-offs, Frameworks, and Emerging Directions

Balancing privacy, accuracy, and fairness requires integrated frameworks. “Towards A Framework for Privacy-Preserving Pedestrian Analysis” [39], a proposed privacy-preserving pedestrian analysis framework using 3D wireframes to anonymize images while retaining pose and context, achieving high utility with low privacy leakage. Abdulaziz in the research [11] developed a three-dimensional benchmark for visual privacy protection, evaluating human perception alongside algorithmic metrics.

Researchers Ardabili et al. discussed policy and technical aspects of AI-enabled surveillance, recommending hybrid approaches combining anonymization with regulatory compliance [5]. Techniques like non-visual sensors reviewed crowd monitoring, emphasizing privacy-preserving to avoid video-based biases [40].

Emerging directions include contextually dependent identification [41], where privacy risks are assessed based on scene context rather than blanket obfuscation. Veltmeijer and Gerritsen [25] examined legal-ethical implications under the EU AI Act, stressing human-in-the-loop for bias mitigation. Differential privacy integration enables human detection in anonymized videos using deep learning, though scalability remains challenging [42].

Table 1: Literature Review Comparison

Reference No.	Objective	Methodology	Limitations
[3]	To develop an accurate real-time pedestrian detection system for video surveillance using deep learning.	Employed a fully convolutional YOLO neural network to process video frames efficiently, focusing on convolutional layers for feature extraction and detection in dynamic scenes.	Limited to RGB video inputs without inherent privacy preservation; vulnerable to environmental factors like lighting variations, and lacks evaluation on privacy-utility trade-offs.
[20]	To create a benchmarking dataset for evaluating privacy-preserving techniques in autonomous driving scenarios, focusing on anonymization of identifiable elements like faces and license plates.	Curated 3,447 driving images with annotations for faces (3,895) and license plates (8,028), including fisheye views; established baselines for de-identification methods like blurring, measuring Average Precision (AP50) for detection post-anonymization.	Dataset is geographically limited to European cities, potentially reducing generalizability to diverse global environments; focuses mainly on vehicular contexts, with challenges in handling small or occluded objects leading to performance drops



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[34]	To assess the trade-off between privacy protection via face anonymization and utility in computer vision tasks like pedestrian detection and segmentation.	Compared anonymization methods (e.g., blurring, latent diffusion) on tasks including keypoint detection and face detection using datasets like COCO; measured privacy via re-identification attack rates and utility via performance metrics like accuracy drop.	High re-identification rates persist across methods despite low recognition, indicating incomplete privacy guarantees; evaluated on limited CV tasks, with potential biases in diverse demographic subgroups not fully explored.
[24]	To evaluate fairness in pedestrian detection models by examining biases related to visual attributes like gender, accessories, and viewpoint.	Leveraged CLIP models for automatic attribute labeling on the ECP dataset; computed Equal Opportunity Difference (EOD) metrics to compare recall rates across subgroups, using bootstrap resampling for validation.	Relies on CLIP's accuracy for labeling, which may introduce errors in attribute detection; focuses on specific attributes, overlooking intersectional biases (e.g., race-gender combinations), and lacks real-time implementation.
[39]	To propose a framework for privacy-preserving pedestrian analysis that balances anonymization with behavioral utility in surveillance.	Reconstructed 3D wireframes from images to remove identifiable features while preserving pose and context; integrated digital inpainting for background consistency, evaluated on datasets like JAAD for privacy leakage and utility metrics.	Computationally intensive for real-time applications due to 3D reconstruction; may lose fine-grained details in complex scenes, and requires high-quality input images to avoid inaccuracies in wireframe extraction.

Gaps in Existing Research – Detailed Study

1. Limited Handling of Variable Resolutions and Small/blocked Pedestrians in Real-Time Detection

The standard YOLO architecture struggles with images of varying resolutions and detecting small, closely grouped pedestrians in low-quality surveillance videos[3]. This gap highlights the need for more adaptive models that maintain accuracy in dense, real-world scenes without architectural overhauls.

Current enhancements like self-attention improve urban detection but do not fully address occlusions in dynamic lighting, suggesting a gap in robust, occlusion-aware real-time systems for surveillance [2], [17].



2. Lack of Diverse, Privacy-Integrated Datasets for Global Generalizability

Currently existing datasets like CityPersons [14] and WiderPerson [15] focus on urban crowds but are geographically biased (e.g., primarily Western cities) and lack built-in privacy annotations, leading to gaps in evaluating models on diverse demographics and environments while preserving PII (personally identifiable information).

The papers “PP4AV: A benchmarking Dataset for Privacy-preserving Autonomous Driving” [20], “CrowdTrack: A Benchmark for Difficult Multiple Pedestrian Tracking in Real Scenarios”, [21], [22], and [23]: Benchmarks like PP4AV and CrowdTrack emphasize anonymization and tracking but are limited to specific contexts (e.g., European driving or dense crowds), with challenges in small/occluded objects and no standardization for multi-modal (e.g., thermal-infrared) privacy evaluations across global scenarios.

3. Dependency on Specific Models and Lack of Black-Box Compatibility in Privacy Techniques

Adversarial obfuscation demands white-box access to detection models, leading to performance declines in cross-model applications (e.g., transitioning from Faster R-CNN to YOLOv8), revealing a gap in the absence of privacy methods adaptable to any model within cloud-based surveillance systems. Encryption-based approaches, such as Verifiable Homomorphic Encryption (VHE) and federated learning, guarantee privacy but introduce computational overhead and rely on trusted environments, highlighting gaps in achieving scalable, black-box integration for dynamic video streams [13], [29], [30].

4. Incomplete Privacy-Utility Trade-Offs and Persistent Re-Identification Risks

Face anonymization techniques such as latent diffusion show a minimal loss in utility but sustain high re-identification rates, which highlights gaps in attaining complete privacy protection without impairing computer vision tasks like detection [34]. These approaches retain essential features but demonstrate 20-30% drops in mean average precision (mAP) during intensive anonymization, with unresolved challenges in handling occluded scenes and achieving precise control over the trade-offs between privacy and performance [6], [7], [8].

5. Overlooked Intersectional Biases and Attribute Imbalances in Fairness Evaluations

CLIP-based labeling uncovers viewpoint biases, for example, reduced detection rates for laterally viewed pedestrians, but it depends on the model's accuracy, thereby neglecting intersectional elements such as race-gender combinations and imbalances in attributes within datasets [24]. Anonymization intensifies disparities, such as elevated errors for underrepresented groups, with gaps in developing bias-aware techniques that incorporate differential privacy (DP) or k-anonymity without compromising demographic predictions [36], [10].

6. Undeveloped Integration with Edge Devices and Dynamic Contexts in Frameworks

The 3D wireframe framework preserves privacy and utility but is computationally intensive, limiting real-time edge use and fine detail retention in complex scenes [39].

Frameworks emphasize benchmarks and context-dependent identification but lack testing in non-stationary cameras (e.g., smart vehicles) and TEE compatibility, suggesting gaps in scalable, hybrid policy-technical solutions for emerging AI regulations [11], [40], [41].



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Problem Statement and Research Questions

Problem Statement

Public surveillance systems are widely used for pedestrian monitoring, traffic control, and public safety management, often relying on computer vision (CV) models trained to detect and track individuals in real-time. Common techniques include face blurring, selective encryption, masking, and full-body anonymization. These methods effectively hide personally identifiable information (PII) and comply with privacy regulations, but often reduce the accuracy of detection and recognition models, especially as privacy intensity increases (e.g., heavier blurring or stronger encryption). Selective encryption and region-based scrambling can conceal sensitive regions while maintaining scene comprehensibility and low computational overhead, but may still degrade model performance for tasks requiring fine-grained features.

Existing research has primarily evaluated these techniques individually, without exploring how hybrid or optimized combinations might balance privacy and accuracy more effectively. Therefore, this study aims to develop and evaluate a hybrid privacy-preserving configuration. For instance, combining light face blurring with region-based masking to identify the most efficient trade-off between privacy protection and model performance. The outcomes will support the development of responsible, privacy-aware computer vision systems that maintain both ethical compliance and technical reliability.

Research Questions

RQ1: How do different privacy-preserving techniques (face blurring, face swapping, and full-body anonymization) affect the accuracy and detection performance of pedestrian detection models trained on public surveillance datasets?

RQ2: How does varying the intensity or level of privacy preservation (e.g., light vs. heavy blurring, partial vs. full anonymization) impact the balance between data privacy and model utility?

RQ3: Can an optimized or hybrid combination of privacy-preserving techniques provide a better balance between privacy protection and model performance compared to individual methods?

Research Objectives and Goals

Main Objective

To evaluate and improve privacy-preserving techniques for pedestrian detection in public surveillance by identifying the optimal balance between privacy protection and model performance, including the development and testing of hybrid or optimized combinations of privacy-preserving methods.

Sub Objectives

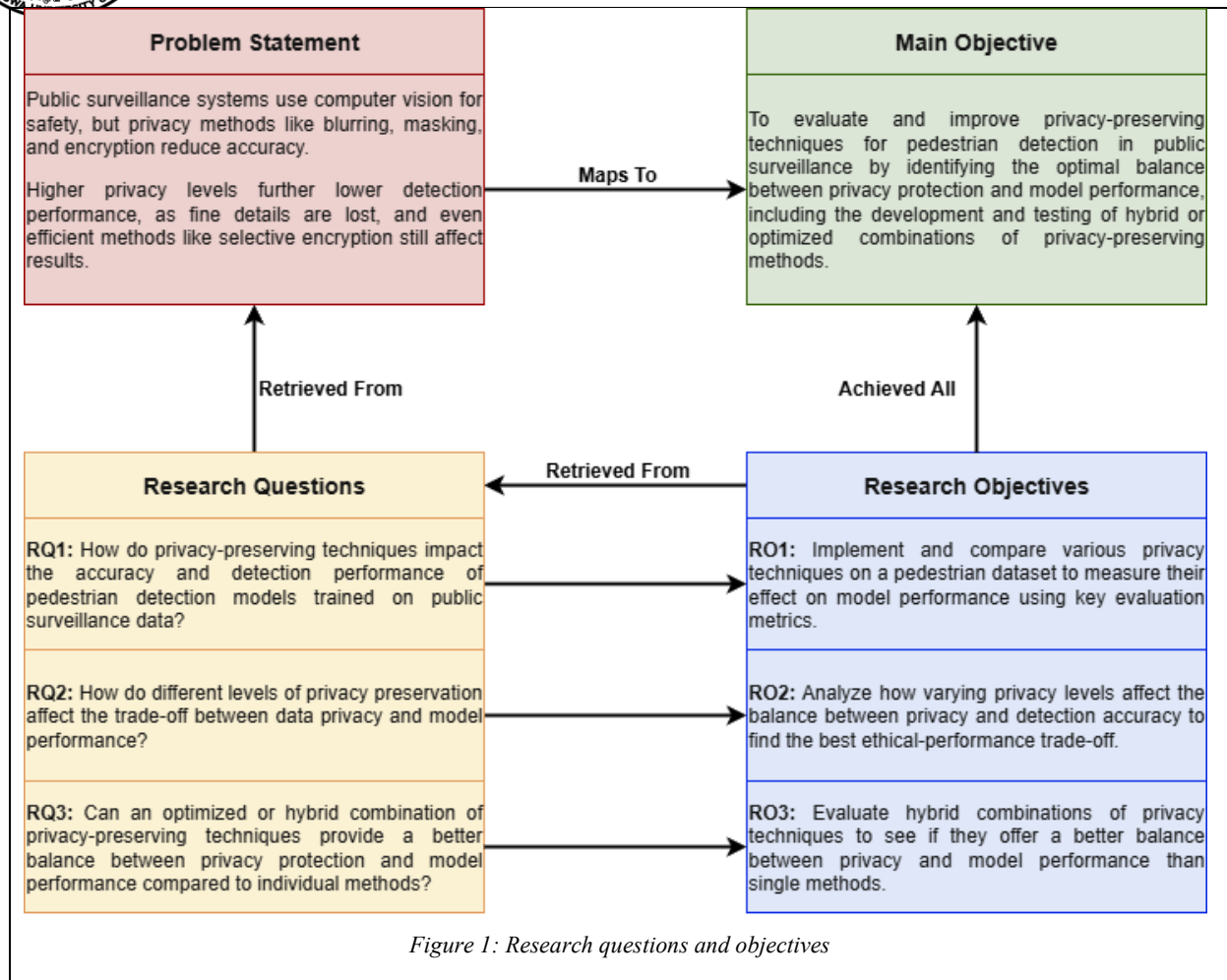
RO1: To implement and compare multiple privacy-preserving techniques like face blurring, face swapping, and full-body anonymization on a pedestrian dataset, and quantitatively evaluate their impact on model performance using metrics such as mAP, precision, recall, and F1-score.

RO2: To examine how different privacy intensity levels influence the trade-off between privacy protection and detection accuracy, and to determine the optimal privacy configuration that balances ethical compliance and model performance.

RO3: To explore and evaluate hybrid or optimized combinations of privacy-preserving techniques to determine whether they provide a better compromise between privacy and model performance compared to individual methods.



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Proposed Research Method

This research plans to adopt a quantitative and experimental research design to systematically evaluate how different privacy-preserving techniques, their intensity levels, and hybrid combinations affect the performance and fairness of pedestrian detection models trained on public surveillance datasets. The overall methodology includes dataset preprocessing, privacy transformation, model training, performance evaluation, and comparative analysis.

1. Experimental Framework

The experimental setup will use publicly available pedestrian detection datasets, including CityPersons, WiderPerson, or a similar dataset, which capture diverse real-world conditions such as lighting, crowd density, and demographic variety. Where applicable, CrowdTrack or KAIST Multispectral datasets will be used for cross-validation and thermal-based privacy comparison.

Each dataset will undergo controlled privacy-preserving transformations to simulate realistic anonymization conditions. The study will focus on three base techniques and their hybrid combinations:

- Face Blurring: Gaussian or mosaic blur at three intensity levels (light, medium, heavy).
- Face Swapping: Identity replacement using DeepPrivacy 2 or a similar generative model.
- Full-Body Anonymization: Semantic segmentation with texture replacement or body masking.



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- Hybrid Configurations: Combinations such as light blurring or partial anonymization or face swapping, or masking, developed to test potential performance improvements.

This transformation process will produce multiple dataset variants for a variety of comparative evaluations across privacy configurations and hybrid settings.

2. Model Training and Configuration

Two commonly used pedestrian detection models will be trained and tested for balanced coverage of accuracy, efficiency, and real-time capability:

- YOLOv8: For high-speed detection and as the baseline benchmark model.
- Faster R-CNN: For high-accuracy, region-based detection performance analysis.

Each model will be trained separately on:

- The original (non-modified) dataset,
- Individually privatized (transformed) datasets (using one technique), and
- Hybrid or optimized privacy configurations.

All models will use consistent hyperparameters and training conditions to ensure comparability. Standard data augmentation and cross-validation procedures will be employed to enhance result reliability.

3. Evaluation Metrics

Model evaluation will be conducted across three major stages:

- Accuracy and Utility Metrics:
 - Mean Average Precision (mAP)
 - Precision, Recall, and F1-Score
 - Inference Speed (Frames Per Second, FPS)
- Fairness and Bias Metrics:
 - Demographic Parity Difference,
 - Equalized Odds, and
 - Subgroup Accuracy Difference (e.g., gender, clothing color, lighting).Fairness evaluation will utilize libraries such as FairLearn or AIF360.
- Privacy–Utility Trade-off Metrics:
 - Quantitative assessment of how increasing privacy intensity or combining techniques affects model accuracy, providing numerical relationships for each configuration.

4. Analysis and Interpretation

Statistical methods such as ANOVA (Analysis of Variance), correlation analysis, and regression modeling will reveal the significance of performance variations induced by differing privacy levels and hybrid configurations, yet challenges remain in fully capturing these effects across diverse scenarios. Trade-off visualizations, including privacy-accuracy-fairness curves, heatmaps, and box plots, will illustrate variable relationships, though gaps persist in standardizing these representations for occluded or complex urban settings. Furthermore, comparisons between individual and hybrid privacy techniques will highlight whether hybrid approaches offer an optimal balance, while subgroup analyses will expose potential fairness enhancements or biases arising from specific transformations, with unresolved issues in addressing intersectional disparities.



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5. Optimization and Framework Development

Experimental outcomes will guide the development of an optimized hybrid privacy configuration that reduces accuracy degradation while preserving acceptable privacy levels, yet limitations arise in scaling this to real-time applications. Additionally, a quantitative privacy–utility model or guideline framework will be proposed to define the interplay among privacy strength, detection accuracy, and fairness outcomes, though gaps remain in integrating these into practical, ethically aligned surveillance AI systems that meet operational demands.

Table 2 : Summary of Methodology

Component	Description
Base Task	Pedestrian detection in public surveillance videos under varying privacy-preserving conditions.
Research Design	Quantitative and experimental study comparing individual, hybrid, and optimized privacy-preserving techniques.
Baseline Models	YOLOv8 (real-time benchmark), Faster R-CNN (accuracy-focused), and SSD-MobileNet (lightweight baseline).
Privacy Techniques	Face blurring, face swapping (using DeepPrivacy 2), and full-body anonymization (segmentation-based masking).
Hybrid Configurations	Combinations of multiple privacy techniques (e.g., light blurring, masking, or swapping and anonymization) to identify optimal privacy–utility balance.
Privacy Intensity Levels	Three transformation levels of light, medium, and heavy were applied consistently across techniques for controlled comparison.
Datasets	Public pedestrian datasets with diverse demographics and environments, including CityPersons, WiderPerson, EuroCity Persons, PP4AV, and KAIST Multispectral.
Evaluation Metrics	Accuracy measurement: mAP, precision, recall, F1-score, FPS. Fairness: demographic parity difference, equalized odds, subgroup accuracy difference. Trade-off: privacy–utility curves.
Statistical Analysis	ANOVA and correlation analysis to assess the effect of privacy levels and hybrid configurations on model performance and fairness.
Implementation Tools	Python, OpenCV, PyTorch/TensorFlow, Scikit-learn, FairLearn, and AIF360 for bias analysis and visualization.



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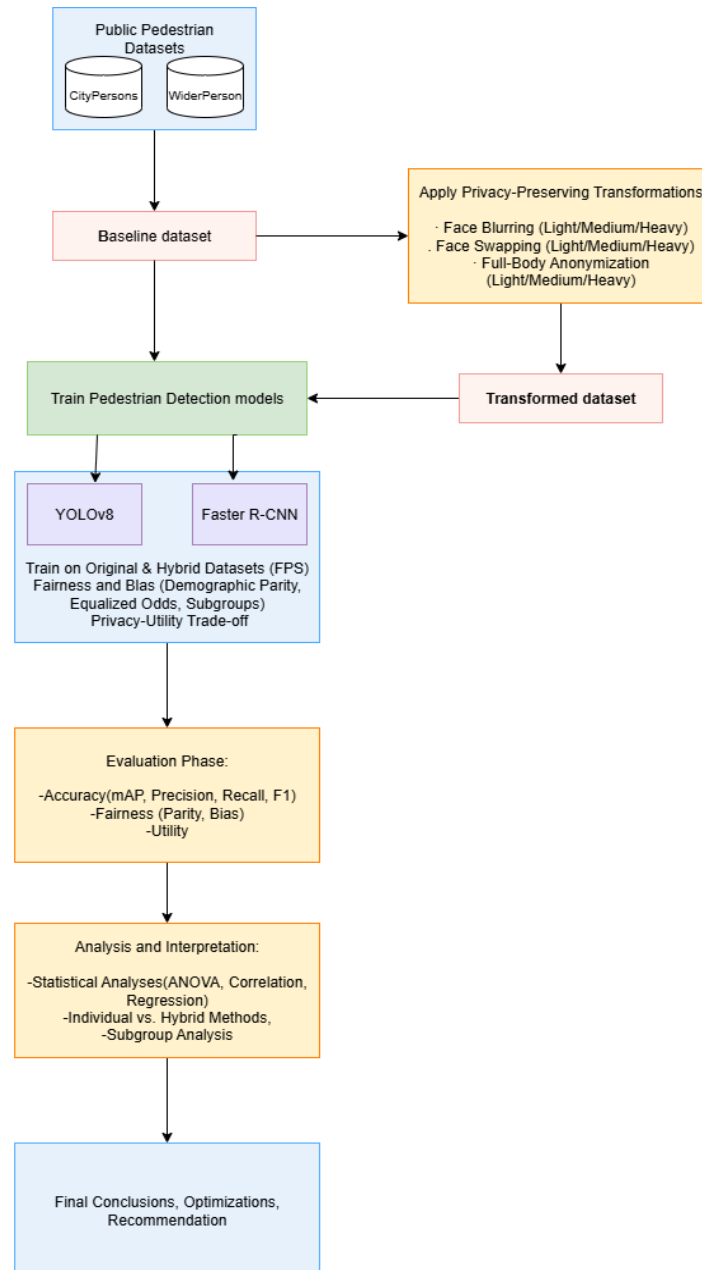


Figure 2: Proposed Research Method Flow



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Expected Outcome

This study is expected to generate both empirical findings and practical contributions to the field of privacy-preserving computer vision, with a focus on improving the trade-off between privacy, accuracy, and fairness in public surveillance systems.

- Results and Comparative Analysis
 - Quantitative evaluation of individual privacy-preserving techniques (face blurring, face swapping, and full-body anonymization) on the accuracy and detection performance of pedestrian detection models.
 - Comparative results show how different privacy intensity levels (light, medium, heavy) affect model utility and detection consistency.
 - Experimental evidence on the impact of hybrid or optimized privacy-preserving configurations, identifying combinations that achieve improved privacy protection with minimal accuracy loss.
- Privacy–Utility Trade-off Insights
 - Development of detailed privacy–utility curves and performance heatmaps illustrating how detection performance varies with privacy intensity and configuration.
 - Statistical analysis identifying the optimal balance point between privacy preservation and model performance, supported by significance testing and correlation measures.
- Fairness and Bias
 - Evaluation of how different privacy-preserving transformations influence fairness across demographic and contextual subgroups, using bias detection metrics such as demographic parity difference and equalized odds.
 - Insights into whether privacy transformations amplify or reduce bias, contributing to a broader understanding of ethical AI practices in surveillance contexts.
- Optimized Hybrid Privacy Configuration
 - A proposed optimized or hybrid privacy-preserving technique that combines multiple methods (e.g., blurring or masking, light anonymization, or selective encryption) to balance privacy and model performance.
 - Documentation of the implementation approach, performance metrics, and reproducibility guidelines for future researchers and practitioners.

Based on these findings, the paper shall present a list of feasible recommendations on privacy procedures that should be chosen regarding accuracy, fairness, and compliance. Reports will be provided in the form of charts, tables, and a suggested privacy-utility-fairness model that will inform further research and be helpful for the improvement of surveillance systems.



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Research Plan and Timetable																											
Table 3: Research Plan																											
Activity	Time Duration																										
	July	August			September			October			November			December			January			February							
Selecting a Supervisor, Research Area, and topic	■	■	■	■																							
Conduct Literature Review				■	■	■	■	■																			
Preparing the Proposal									■	■																	
Proposal Submission and Proposal Defense										■	■																
Collecting the Dataset											■	■	■	■													
Analyzing the Data												■	■	■													
Choosing and Implementing the Model													■	■	■	■											
Rerun the models by applying privacy-preserving techniques															■	■	■										
Model Evaluation and documenting the findings																	■	■									
Conclusion and Recommendation																			■	■							
Preparing final report and thesis																					■	■					
Final Research Submission																							■	■			
Final research defense																									■	■	

Proposed Budget	
The Internet fees and publication costs are the expenses for the research.	



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References

- [1] G. Dobler, J. Vani, and T. T. L. Dam, "Patterns of urban foot traffic dynamics," *Comput. Environ. Urban Syst.*, vol. 89, 2021, doi: 10.1016/j.compenvurbsys.2021.101674.
- [2] H. Sui, H. Han, Y. Cui, M. Yang, and B. Pei, "FA-YOLO: A Pedestrian Detection Algorithm with Feature Enhancement and Adaptive Sparse Self-Attention," *Electronics*, vol. 14, no. 9, p. 1713, Apr. 2025, doi: 10.3390/electronics14091713.
- [3] V. V. Molchanov, B. V. Vishnyakov, Y. V. Vizilter, O. V. Vishnyakova, and V. A. Knyaz, "Pedestrian detection in video surveillance using fully convolutional YOLO neural network," J. Beyerer and F. Puente León, Eds., Jun. 2017, p. 103340Q. doi: 10.1117/12.2270326.
- [4] J. Chavat, S. Nesmachnow, A. Tchernykh, and V. Shepelev, "Active Safety System for Urban Environments with Detecting Harmful Pedestrian Movement Patterns Using Computational Intelligence," *Appl. Sci.*, vol. 10, no. 24, p. 9021, Dec. 2020, doi: 10.3390/app10249021.
- [5] B. R. Ardabili *et al.*, "Understanding Policy and Technical Aspects of AI-enabled Smart Video Surveillance to Address Public Safety," *Comput. Urban Sci.*, vol. 3, no. 1, p. 21, May 2023, doi: 10.1007/s43762-023-00097-8.
- [6] A. S. Abdelrahman, M. Abdel-Aty, and D. Wang, "Video-to-Text Pedestrian Monitoring (VTPM): Leveraging Computer Vision and Large Language Models for Privacy-Preserve Pedestrian Activity Monitoring at Intersections," Aug. 29, 2024. doi: 10.21203/rs.3.rs-4954456/v1.
- [7] S. C. Triess, T. Leitritz, and C. Jauch, "Exploring AI-based Anonymization of Industrial Image and Video Data in the Context of Feature Preservation," May 2024, [Online]. Available: <http://arxiv.org/abs/2405.19173>
- [8] H. Hukkelås and F. Lindseth, "Does Image Anonymization Impact Computer Vision Training?," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, IEEE, Jun. 2023, pp. 140–150. doi: 10.1109/CVPRW59228.2023.00019.
- [9] M. Yaghini, P. Liu, F. Boenisch, and N. Papernot, "Learning with Impartiality to Walk on the Pareto Frontier of Fairness, Privacy, and Utility," Feb. 2023, [Online]. Available: <http://arxiv.org/abs/2302.09183>
- [10] S. Noiret, S. Ravi, M. Kampel, and F. Florez-Revuelta, "Fairly Private: Investigating The Fairness of Visual Privacy Preservation Algorithms," 2023, [Online]. Available: <http://arxiv.org/abs/2301.05012>
- [11] S. Abdulaziz, G. D'Amicantonio, and E. Bondarev, "Evaluation of Human Visual Privacy Protection: A Three-Dimensional Framework and Benchmark Dataset," Jul. 2025, [Online]. Available: <http://arxiv.org/abs/2507.13981>
- [12] W. Paul, P. Mathew, F. Alajaji, and P. Burlina, "Evaluating Trade-offs in Computer Vision Between Attribute Privacy, Fairness and Utility," Feb. 2023, [Online]. Available: <http://arxiv.org/abs/2302.07917>
- [13] H. Yang, Q. Zhou, J. Ni, H. Li, and X. Shen, "Accurate Image-Based Pedestrian Detection With Privacy Preservation," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 14494–14509, Dec. 2020, doi: 10.1109/TVT.2020.3043203.
- [14] S. Zhang, R. Benenson, and B. Schiele, "CityPersons: A diverse dataset for pedestrian detection," in *Proceedings - 30th IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017*, 2017. doi: 10.1109/CVPR.2017.474.
- [15] S. Zhang, Y. Xie, J. Wan, H. Xia, S. Z. Li, and G. Guo, "WiderPerson: A Diverse Dataset for Dense Pedestrian Detection in the Wild," *IEEE Trans. Multimed.*, vol. 22, no. 2, 2020, doi: 10.1109/TMM.2019.2929005.
- [16] E. Bentafat, M. M. Rathore, and S. Bakiras, "Towards real-time privacy-preserving video surveillance," *Comput. Commun.*, vol. 180, pp. 97–108, Dec. 2021, doi: 10.1016/j.comcom.2021.09.009.
- [17] M. P. Philipsen, M. B. Jensen, M. M. Trivedi, A. Mogelmose, and T. B. Moeslund, "Ongoing work on traffic lights: Detection and evaluation," in *2015 12th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, IEEE, Aug. 2015, pp. 1–6. doi: 10.1109/AVSS.2015.7301730.
- [18] M. Fanfani, M. Marulli, and P. Nesi, "Evaluation of Geometric and Photometric Data Augmentation for Pedestrian Detection with Thermal Cameras," 2024, pp. 353–366. doi: 10.1007/978-3-031-65318-6_24.
- [19] R. Shen, N. Inoue, and K. Shinoda, "Text-Guided Object Detector for Multi-modal Video Question Answering," in *2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, IEEE, Jan. 2023, pp. 1032–1042. doi: 10.1109/WACV56688.2023.00109.



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- [20] L. Trinh *et al.*, “PP4AV: A benchmarking Dataset for Privacy-preserving Autonomous Driving,” in *2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, IEEE, Jan. 2023, pp. 1206–1215. doi: 10.1109/WACV56688.2023.00126.
- [21] T. Fu, Y. Chen, Z. Chen, M. Zhao, B. Li, and X. Xue, “CrowdTrack: A Benchmark for Difficult Multiple Pedestrian Tracking in Real Scenarios,” Jul. 2025, [Online]. Available: <http://arxiv.org/abs/2507.02479>
- [22] S. Hwang, J. Park, N. Kim, Y. Choi, and I. S. Kweon, “Multispectral pedestrian detection: Benchmark dataset and baseline,” in *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, Jun. 2015, pp. 1037–1045. doi: 10.1109/CVPR.2015.7298706.
- [23] M. Braun, S. Krebs, F. Flohr, and D. M. Gavrilu, “EuroCity Persons: A Novel Benchmark for Person Detection in Traffic Scenes,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 8, pp. 1844–1861, Aug. 2019, doi: 10.1109/TPAMI.2019.2897684.
- [24] M. Khoshkhdahan, N. Kj  r, and F. B. Flohr, “FAIR-PED: Fairness Evaluation in Pedestrian Detection Using CLIP,” in *2025 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, Jun. 2025, pp. 1504–1509. doi: 10.1109/IV64158.2025.11097691.
- [25] E. Veltmeijer and C. Gerritsen, “Legal and ethical implications of AI-based crowd analysis: the AI Act and beyond,” *AI Ethics*, vol. 5, no. 3, pp. 3173–3183, Jun. 2025, doi: 10.1007/s43681-024-00644-x.
- [26] F. Dufaux and T. Ebrahimi, “Scrambling for Privacy Protection in Video Surveillance Systems,” *IEEE Trans. Circuits Syst. Video Technol.*, vol. 18, no. 8, pp. 1168–1174, Aug. 2008, doi: 10.1109/TCSVT.2008.928225.
- [27] K. M. Hosny, M. A. Zaki, H. M. Hamza, M. M. Fouda, and N. A. Lashin, “Privacy Protection in Surveillance Videos Using Block Scrambling-Based Encryption and DCNN-Based Face Detection,” *IEEE Access*, vol. 10, pp. 106750–106769, 2022, doi: 10.1109/ACCESS.2022.3211657.
- [28] S. De Coninck, W.-C. Wang, S. Leroux, and P. Simoons, “Privacy-preserving visual analysis: training video obfuscation models without sensitive labels,” *Appl. Intell.*, vol. 54, no. 8, pp. 6041–6052, Apr. 2024, doi: 10.1007/s10489-024-05489-9.
- [29] X. Tian, P. Zheng, and J. Huang, “Robust Privacy-Preserving Motion Detection and Object Tracking in Encrypted Streaming Video,” *IEEE Trans. Inf. Forensics Secur.*, vol. 16, pp. 5381–5396, 2021, doi: 10.1109/TIFS.2021.3128817.
- [30] M. Nabil, A. Sherif, M. Mahmoud, W. Alsmay, and M. Alsabaan, “Accurate and Privacy-Preserving Person Localization Using Federated-Learning and the Camera Surveillance Systems of Public Places,” *IEEE Access*, vol. 10, pp. 109894–109907, 2022, doi: 10.1109/ACCESS.2022.3214227.
- [31] H. Xu, Z. Cai, D. Takabi, and W. Li, “Audio-Visual Autoencoding for Privacy-Preserving Video Streaming,” *IEEE Internet Things J.*, vol. 9, no. 3, pp. 1749–1761, Feb. 2022, doi: 10.1109/JIOT.2021.3089080.
- [32] H. Lee, M. U. Kim, Y. Kim, H. Lyu, and H. J. Yang, “Development of a Privacy-Preserving UAV System with Deep Learning-Based Face Anonymization,” *IEEE Access*, vol. 9, 2021, doi: 10.1109/ACCESS.2021.3113186.
- [33] H. Li and J. T. Y. Lo, “A review on the use of top-view surveillance videos for pedestrian detection, tracking and behavior recognition across public spaces,” *Accid. Anal. Prev.*, vol. 215, p. 107986, Jun. 2025, doi: 10.1016/j.aap.2025.107986.
- [34] R. Stenger, S. Busse, J. Sander, T. Eisenbarth, and S. Fudickar, “Evaluating the Impact of Face Anonymization Methods on Computer Vision Tasks: A Trade-Off Between Privacy and Utility,” *IEEE Access*, vol. 13, pp. 11070–11079, 2025, doi: 10.1109/ACCESS.2024.3519441.
- [35] X. Li, Z. Chen, J. M. Zhang, F. Sarro, Y. Zhang, and X. Liu, “Bias Behind the Wheel: Fairness Testing of Autonomous Driving Systems,” Oct. 2024, [Online]. Available: <http://arxiv.org/abs/2308.02935>
- [36] J. Buolamwini and T. Gebu, “Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification,” in *Proceedings of Machine Learning Research*, 2018.
- [37] J. Zhao, Y. Chen, and W. Zhang, “Differential Privacy Preservation in Deep Learning: Challenges, Opportunities and Solutions,” *IEEE Access*, vol. 7, 2019, doi: 10.1109/ACCESS.2019.2909559.
- [38] L. Maris, Y. Matsuda, and K. Yasumoto, “Differential Privacy and K-anonymity for Pedestrian Image Data: Impact on Cross-camera Person Re-identification & Demographic Predictions,” *ACM Trans. Cyber-Physical Syst.*, Jun. 2025, doi: 10.1145/3743680.
- [39] A. Kunchala, M. Bourroche, and B. Schoen-Phelan, “Towards A Framework for Privacy-Preserving Pedestrian Analysis,” in *2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, IEEE, Jan. 2023, pp. 4359–4369. doi: 10.1109/WACV56688.2023.00435.



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| [40] | U. Singh, J.-F. Determe, F. Horlin, and P. De Doncker, "Crowd Monitoring: State-of-the-Art and Future Directions," <i>IETE Tech. Rev.</i> , vol. 38, no. 6, pp. 578–594, Nov. 2021, doi: 10.1080/02564602.2020.1803152. |
| [41] | M. Burdon, T. Cohen, J. Buckley, and M. Milford, "From object obfuscation to contextually-dependent identification: enhancing automated privacy protection in street-level image platforms (SLIPs)," <i>Inf. Commun. Technol. Law</i> , vol. 33, no. 2, pp. 198–221, May 2024, doi: 10.1080/13600834.2024.2321052. |
| [42] | M. J. Yousuf, N. Kanwal, M. S. Ansari, M. Asghar, and B. Lee, "Deep Learning based Human Detection in Privacy-Preserved Surveillance Videos," Jul. 2022. doi: 10.14236/ewic/HCI2022.33. |

Any Other Relevant Information